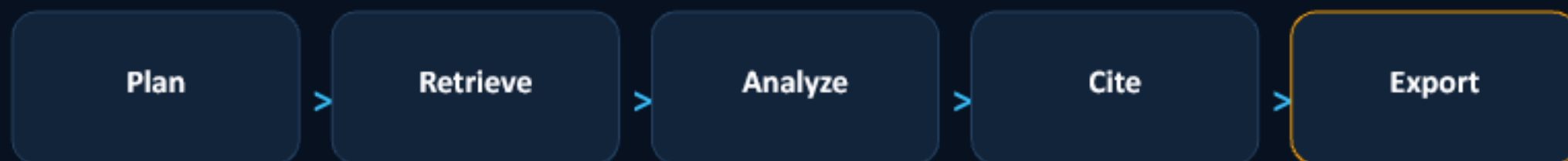


CapitalLens AI

Autonomous Financial Research & Productivity Agent

Finance track submission by Sai-Vivek17



8

Agent modules

5

Demo companies

Yes

Runs without API
keys

The Research Problem

Public-company research is high-stakes knowledge work, but the first pass is usually manual, fragmented, and hard to audit.

- Analysts jump between market data, filings, news, notes, and spreadsheets.
- LLM-only summaries sound confident but often lose source fidelity.
- Students and founders need a repeatable research workflow, not a generic chatbot.
- Finance teams need citations, risk flags, and exports that can be reviewed.

**Cited memo in
minutes**

Goal

**Safe research,
not advice**

Constraint

Target Users

CapitalLens turns scattered public-company data into a structured research memo that is fast to read, easy to cite, and safe to demo.

Equity analysts

Screen and brief
companies

Students

Learn with cited
evidence

Founders

Research competitors

Finance teams

Monitor watchlists

Real utility: repeatable, reviewable, exportable

Finance Track Fit

The system maps directly to finance workflows: company overview, financial health, recent events, risk analysis, valuation context, and watchlist monitoring.

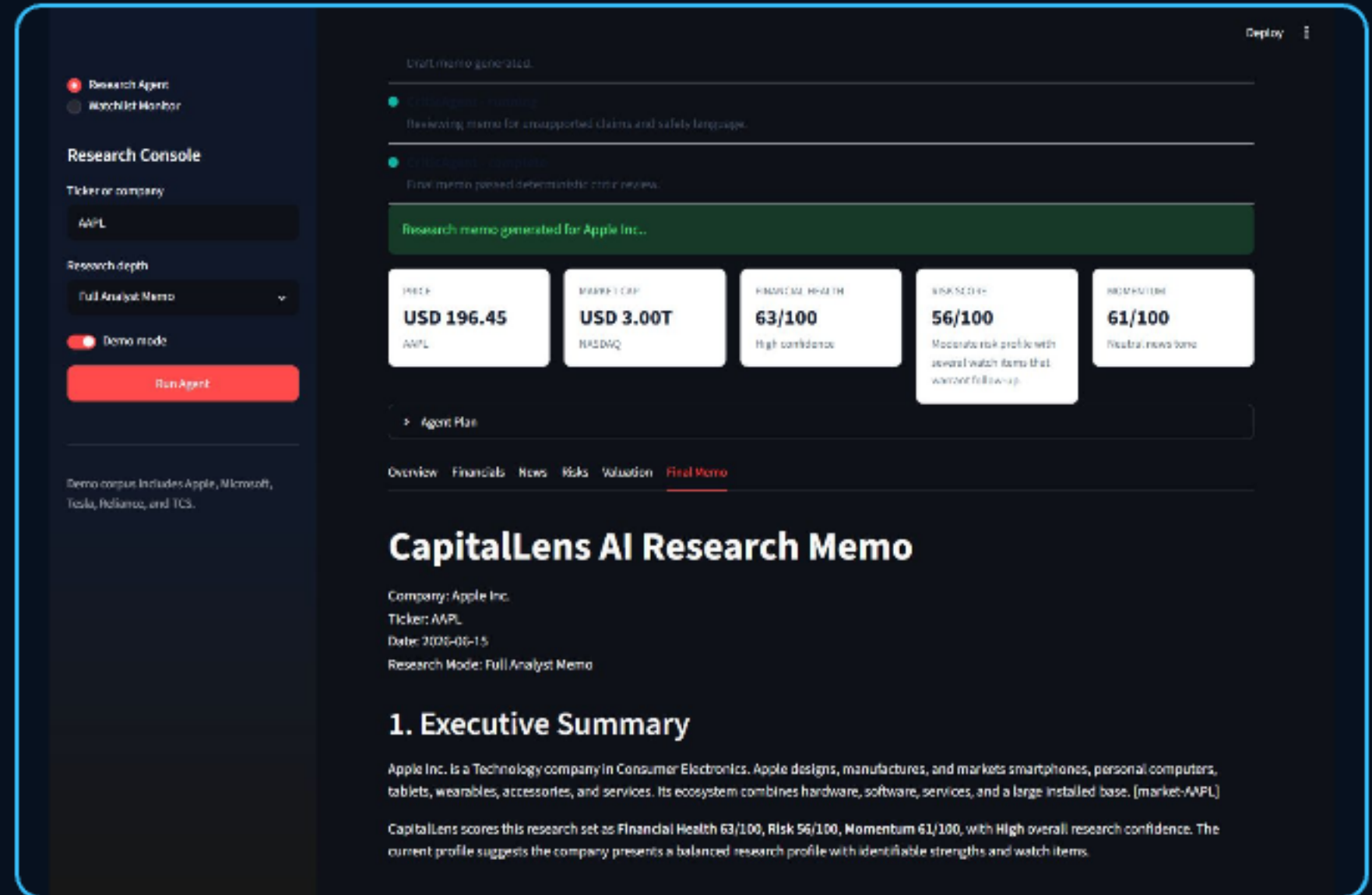


- Accepts US and India tickers such as AAPL, MSFT, TSLA, RELIANCE.NS, and TCS.NS.
- Uses deterministic demo data when live data or API keys are unavailable.
- Preserves educational and research-only safety boundaries throughout the memo.

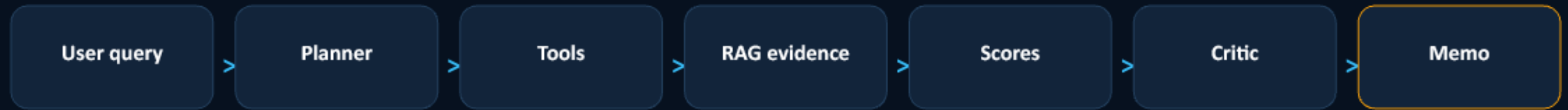
Working Product Snapshot

The Streamlit console exposes the complete agent workflow: settings, progress timeline, score cards, charts, tabs, final memo, and exports.

- Sidebar: ticker, mode, demo toggle
- Tabs: overview, financials, news, risks, valuation, memo
- Downloads: Markdown and PDF



Autonomous Core Workflow



The user selects Quick Scan, Full Analyst Memo, or Risk-First Review. The agent decomposes the task, gathers data, retrieves evidence, scores risk, checks citation support, then produces a professional memo.

3

Modes

**Structured
JSON + memo**

Outputs

**Markdown /
PDF**

Export

Multi-Agent Architecture

CapitalLens is not a single prompt wrapper. Each module owns a distinct research responsibility and exchanges typed Pydantic outputs.

Planner

Market Data

News

Filing RAG

Risk

Valuation

Report

Critic

Orchestrated by CapitalLens research engine

Durable Agent Execution

Phase 2 adds run IDs, persisted event logs, retry-safe execution, and status transitions so every research run is observable and reviewable.



- DurableAgentExecutor wraps each specialized agent with configured retries.
- GET /runs/{run_id}/events exposes planner and agent progress for UI observability.
- Failures are explicit, logged, and recoverable in demo mode.

Structured Outputs

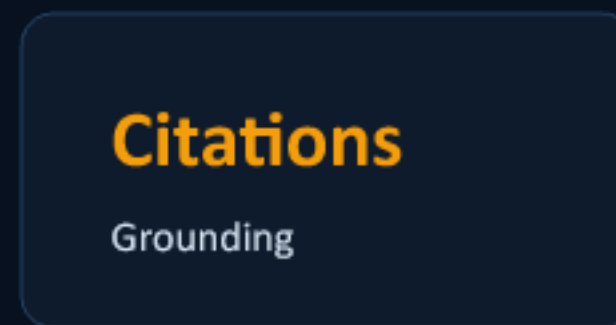
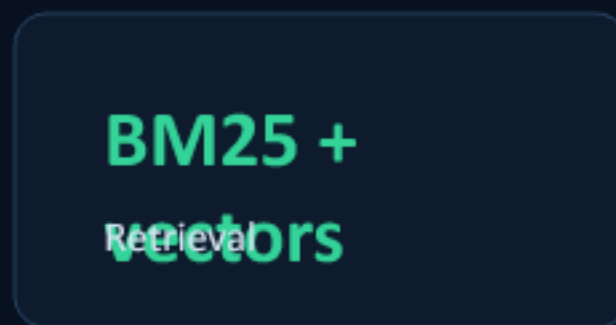
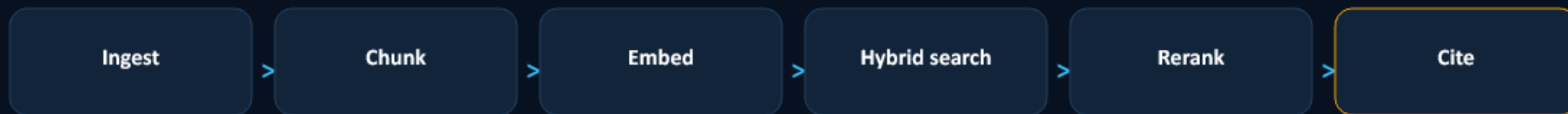
Pydantic schemas make every agent output predictable: market metrics, news events, evidence chunks, risk entries, valuation ratios, report scores, and final memo metadata.



- Reduces brittle prompt parsing and UI breakage.
- Allows tests to assert specific finance and RAG behavior.
- Makes missing data explicit instead of hallucinated.

Advanced RAG Pipeline

The filing agent builds a persistent local retrieval layer over sample filings and best-effort SEC text, then retrieves evidence for business model, risks, revenue, competition, debt, and management commentary.



Source Fidelity Controls

The critic and citation audit layer checks whether the memo is grounded in retrieved evidence before the final report is shown.

- Citation coverage is measured and surfaced in report metadata.
- Unsupported retrieval topics are flagged for review.
- Weak claims are softened instead of overstated.
- The final memo includes sources and a clear research disclaimer.

Faithfulness first

Useful finance
output only
matters if judges
can trace the
claims.

Data Sources

CapitalLens combines live-ready data adapters with deterministic fallback data so demos remain reliable during judging.

Market

yfinance when available; demo metrics when offline

Filings

SEC/EDGAR best effort plus sample filing corpus

News

Optional provider through env variable; fallback events

Storage

SQLite watchlist and persistent RAG index

Risk and Valuation Intelligence

The analysis layer produces severity-scored risks, a valuation snapshot, peer context where available, and professional bull/bear framing.

Risk matrix

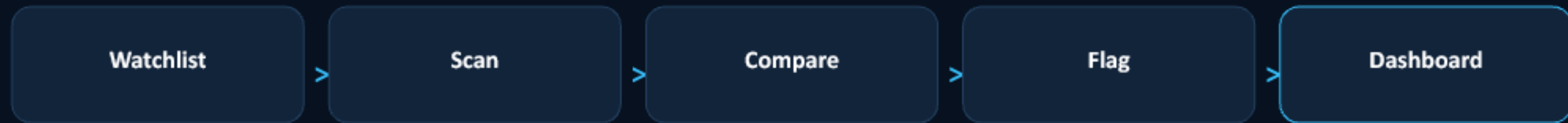
- Financial
- Business
- Market
- Regulatory
- Execution

Valuation snapshot

- P/E
- P/S
- EV/EBITDA
- Peer comparison
- Unavailable data disclosed

Watchlist Automation

Users can add multiple tickers and run a monitor that flags unusual changes or emerging risks with prior scan context stored locally.



- Large price movement and high volatility
- Negative news or risk-score increase
- Weak financial trend or deteriorating signals

Demo-Friendly UX

The frontend is designed for a live hackathon demo: a serious finance console, not a chat box.

- Sidebar settings for ticker, research depth, and demo mode.
- Progress timeline shows each autonomous agent step.
- Tabs separate overview, financials, news, risks, valuation, and memo.
- Charts, score cards, risk matrix, Markdown download, and PDF export.

Judges can run it locally in under 5 minutes

Safety and Compliance

CapitalLens avoids direct trading instructions and frames outputs as research support for education and review.

- No direct buy, sell, or trade recommendations.
- Missing metrics are marked unavailable instead of guessed.
- Citations and source lists are included when evidence exists.
- Final disclaimer states the system is not financial, legal, tax, or trading advice.

Production Readiness

The repository includes the engineering scaffolding judges expect from a serious full-stack Agentic RAG project.

FastAPI backend

Streamlit frontend

Typed schemas

SQLite storage

Docker + Compose

CI tests

.env.example

Sample reports

Verification target: pytest, backend health, demo AAPL research request, Streamlit launch, and watchlist scan.

Demo Reliability

Hackathon judging environments are unpredictable. Demo mode ensures the product still works even without OpenAI, NewsAPI, or live market connectivity.



9 passed

Tests

Deterministic

Fallback

Repeatable

Judging

Business Impact

CapitalLens compresses the first-pass company research workflow while keeping evidence, risk, and safety visible.

- Analysts get a credible starting memo faster.
- Students learn how to connect claims to sources.
- Founders can research competitors and public comparables.
- Finance teams can monitor watchlists and review alerts.

Clear utility + real architecture + reliable demo

Roadmap and Submission Ask

The current project is a working local product. The next phase turns it into a hosted, team-ready finance research platform.

- Managed vector DB or Chroma Cloud with source freshness scoring.
- Deeper SEC/XBRL ingestion, richer peer discovery, and DCF scenarios.
- Auth, scheduled scans, Slack/email alerts, and team workspaces.
- Observability dashboards, benchmark suites, and faithfulness evaluation.

CapitalLens AI: a finance-track Agentic RAG system built for real-world research productivity.